Jason (Doohyun) Lee

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Nationality: U.S. & South Korea Dual Citizen

Education

Georgia Institute of Technology — Aug. 2026 – Present
Ph.D. Student, Electrical and Computer Engineering — *Atlanta, GA*

Yonsei University — Mar. 2020 – Aug. 2026
B.S., Electrical and Electronic Engineering — *Seoul, South Korea*
Advisor: Ki Jun Yu
GPA: 3.71 / 4.0

Research Interests

Electronic Materials & Thin-Film Devices	1. Nano / Micro scale semiconductor fabrication technologies and thin-film devices 2. CMOS-compatible processing of single-crystalline silicon nanomembranes
Semiconductor Packaging & Integration	1. 2.5D/3D heterogeneous integration for next-generation electronic systems 2. Glass substrate and interposer technologies for advanced packaging

Research Experience

Georgia Institute of Technology — Sep. 2026 – Present
Graduate Research Assistant — *Atlanta, GA*

- Conducting research on advanced semiconductor packaging and 2.5D/3D heterogeneous integration for next-generation electronic systems. Currently focusing on glass-core substrate technologies and system-technology co-optimization (STCO) for advanced packaging

Yonsei University — Jun. 2022 – Aug. 2026
Undergraduate Research Assistant — *Seoul, South Korea*
Functional Bio-Integrated Electronics and Energy Management Lab
(PI: Ki Jun Yu)

- Conducted research on electronic materials and CMOS-compatible manufacturing of silicon membranes for bio-integrated electronics such as wearable e-skins, thin-film silicon transistors, and neural interfaces
- Contributed to setting up semiconductor processing equipment, drafting research grant proposals, and facilitating interdisciplinary collaboration across electrical engineering, mechanical engineering, materials science, and bioengineering

ASEM — May 2025 – Aug. 2026
Co-Executive Director — *Hwaseong, South Korea*
Thin-film VCSEL Tech Startup

- Contributed to the development of GaAs-based thin-film 940 nm vertical-cavity surface-emitting lasers (VCSELs) grown with a metal–organic chemical vapor deposition system
- Secured $200,000 in research funding for optoelectronic devices enabling LiDAR sensing

Yonsei University — Jul. 2025 – Dec. 2025
Undergraduate Semiconductor Research Project — *Seoul, South Korea*
Semiconductor Engineering Lab
(PI: Ilgu Yun)

- Carried out a research project on the following theme: “Ferroelectric field-effect transistors with high-k gate stacks: strategies for memory window improvement in HZO-based MFIS FeFETs”
- Performed device-level modeling and material/structural optimization using Silvaco TCAD to improve the memory window of FeFETs

Research Portfolio (Link to Google Scholar)

Publications

†: co-first authors
*: corresponding authors

1. Taemin Kim†, Kyubeen Kim†, In Sik Min†, Ankan Dutta, Giyoung Kim, Jongchan Han, Jongwoon Shin, Jeong Eun Ju, **Doohyun J. Lee**, Heekyeong Choi, Hokeon Lee, Myeongki Cho, Jaejin Park, Kyowon Kang, Mingyu Sang, Huanyu Cheng, Ju Young Lee*, Ki Jun Yu*, "Temperature-compensated silicon nanomembrane gauges for monitoring lithium-ion battery swelling", *Device*, 2026.

2. Junseok Lee†, Yena Lee†, Jongwoon Shin†, **Doohyun J. Lee†**, Hyosang Yoon, Ju Young Lee*, Ki Jun Yu*, Mingyu Sang*, "Flexible and stretchable bioelectronics for sports wearable monitoring: Materials, Device Engineering, and Intelligent Wearable Platforms", *Biosensors and Bioelectronics,* 2026.

3. Junseok Lee†, Yena Lee†, Hanbi Woo†, Minji Hong, **Doohyun J. Lee**, Mingyu Sang*, "Advanced silicon nanomembrane based bioelectronics for flexible and stretchable implantable systems", *Biomedical Engineering Letters*, 2026.

4. Mingyu Sang†, Kyubeen Kim†, **Doohyun J. Lee†**, Young Uk Cho*, Jung Woo Lee*, Ki Jun Yu*, "Technical roadmap of ultra-thin crystalline silicon-based bioelectronics", *International Journal of Extreme Manufacturing*, 2025.

5. Kyubeen Kim†, Jung-Hoon Hong†, Kyubin Bae†, Kyounghun Lee, **Doohyun J. Lee**, Haozhe Zhang, Junsu Park, Mingyu Sang, Jeong Eun Ju, Young Uk Cho, Kyowon Kang, Suah Jung, Baoxing Xu, Jongbaeg Kim*, Ki Jun Yu*, "Extremely durable electrical impedance tomography-based soft and ultra-thin wearable e-skin for three-dimensional tactile interface", *Science Advances*, 2024.

6. Kyowon Kang†, Kiho Kim†, Junhyeong Baek, **Doohyun J. Lee**, Ki Jun Yu*, "Biomimic and bioinspired soft neuromorphic tactile sensory system", *Applied Physics Reviews*, 2024.

7. Kyubeen Kim†, **Doohyun J. Lee†**, Sang Hoon Park†, Jongwoon Shin†, …, Mingyu Sang*, Ju Young Lee*, Ki Jun Yu*, "Laser-induced gold doping of single-crystalline silicon for wearable core temperature sensing", *Science (submitted)*.

8. Jongwoon Shin†, **Doohyun J. Lee†**, …, Ki Jun Yu*, "A micro-structured flexible gas sensor based on gold nanoparticle-decorated silicon nanomembrane for hazardous gas detection", *(in preparation, co-first-authored)*.

9. Ju Young Lee†, **Doohyun J. Lee†**, …, Ki Jun Yu*, "Fully implantable photodiode for optoelectronic neural modulation with infrared light", *(in preparation, co-first-authored)*.

10. Jongwoon Shin†, …, **Doohyun J. Lee**, …, Ki Jun Yu*, "rGO-decorated silicon nanomembrane for hazardous gas detection", *(in preparation, co-authored)*.

11. Ju Young Lee†, …, **Doohyun J. Lee**, …, Ki Jun Yu*, "LED-integrated flexible neural probe for simultaneous stimulation and recording of electrophysiological signals", *(in preparation, co-authored)*.

Conferences

1. **Doohyun J. Lee**, Kyubeen Kim, Ju Young Lee, Mingyu Sang*, Ki Jun Yu*, "Ultrathin gold-doped silicon nanomembrane sensor for real-time brain surface temperature monitoring", *Fall Conference of the Korean Society of Medical & Biological Engineering (KOSOMBE)*, 2025. *[poster]* ***[best poster award]***
2. **Doohyun J. Lee**, Jongwoon Shin, Yelynn Ryu, Wooyoung Lee*, Ki Jun Yu*, "A micro-structured flexible gas sensor based on gold nanoparticle-decorated silicon nanomembrane for hazardous gas detection", *Fall Conference of Materials Research Society of Korea (MRS-K)*, 2025. *[oral presentation]*
3. **Doohyun J. Lee**, Kyubeen Kim, Ju Young Lee, Mingyu Sang*, Ki Jun Yu*, "Ultrathin gold-doped silicon nanomembrane sensor for real-time brain surface temperature monitoring", *Fall Conference of Korean Sensors Society (KSS)*, 2025. *[poster]*
4. Junseok Lee, **Doohyun J. Lee**, Ki Jun Yu*, Mingyu Sang*, "Wearable sweat sensor with a porous active thermal layer for enhanced signal repolarization and amplification", *Fall Conference of Korean Sensors Society (KSS)*, 2025. *[poster]*
5. Junseok Lee, **Doohyun J. Lee**, Hyosang Yoon, Ki Jun Yu*, Mingyu Sang*, "Effect of temperature on the performance of PEDOT:PSS and graphene oxide electrodes for a potentiometric NaCl concentration sensor", *Fall Conference of the Korean Society of Medical & Biological Engineering (KOSOMBE)*, 2025. *[poster]*

Awards and Grants

Georgia Tech ECE ON Semiconductor Fellowship ····· Sep. 2026
Georgia Institute of Technology
Selected by the School of Electrical and Computer Engineering for a Ph.D. fellowship in the Fall 2026 cohort

DoD SCALE Fellow ····· Sep. 2026
Scalable Asymmetric Lifecycle Engagement (SCALE)
Heterogeneous Integration/Advanced Packaging vertical of a national workforce program in microelectronics

Best Poster Award ····· Nov. 2025
Korean Society of Medical & Biological Engineering (KOSOMBE)
Recognition as an outstanding poster presenter at the 2025 Fall Conference of KOSOMBE

Next-Generation Engineer Scholar Award ····· Sep. 2025
Institute for Promotion of Engineering and Science of Korea
Nation-wide honor awarded to top 51 engineering students for outstanding academic and research performance

Honor Award ····· Aug. 2022
Yonsei University
Academic excellency of upper 10% GPA in the department

Yonsei Student Scholarship ····· Aug. 2022, Jun. 2025
Yonsei University
Merit-based scholarship for students with outstanding academic performance

Semiconductor Track Scholarship ····· Jan. 2022 – Feb. 2023
South Korean Ministry of Trade, Industry and Energy
Merit-based scholarship for distinguished students in the field of semiconductor engineering

Media Coverage

Online Links to Research Spotlights

Link 1 (Department News)
Link 2 (Department News)
Link 3 (LinkedIn)
Link 4 (LinkedIn)

Academic Service

Peer Reviewing of Scientific Journals

- ACS Sensors
- Advanced Science

Technical Skills

Hardware Skill Sets	CMOS microfabrication (doping, thermal processing, thin-film deposition), Layer transferring of silicon nanomembranes for flexible electronics, Material and device characterization
Software Skill Sets	TCAD device simulation (Silvaco), Multiphysics simulation (Ansys HFSS, Lumerical, Keysight ADS), Photonic IC layout and post-layout simulation with PDK (gdsfactory, SAX, KLayout DRC)

References

Jeehwan Kim
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Dept. of Mechanical Engineering
Massachusetts Institute of Technology
Email: jeehwan@mit.edu

Ki Jun Yu
Associate Professor
Dept. of Electrical and Electronic Engineering
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Mingyu Sang
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Hyun S. Kum
Assistant Professor
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